# Anbu

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## Links

**Overview:** github.com/anbu-05 - landing page summarizing my work
**Full Portfolio:** github.com/anbu-05/Portfolio - detailed write-ups of projects and PCB designs

## Skills and Education

**Firmware:** C, C++, Python, Embedded C, FreeRTOS, STM32 Cube HAL, ESP-IDF, MQTT, PID control
**Digital Design:** Verilog, SystemVerilog, RTL design, FPGA design, AXI4-Lite, RISC-V assembly
**Hardware:** STM32, ESP32, Siemens S7-1500 PLC, PCB design, schematic capture, layout, routing, SMD soldering
**Tools:** Git, GitHub, QuestaSim, Cadence Xcelium, Altium Designer, KiCad, RISC-V GCC, TIA Portal, S7-PLCSIM

**Vellore Institute of Technology, Vellore** | **B.Tech EE.VLSI** *Aug'23 – present | CGPA: 7.10*
**Kamala Niketan Montessori School, Trichy** | **Higher Secondary Education (12th)** *2023 | Board: 93.8%*

## Projects | portfolio

**Turtlebot Controller Board** | **DFM & Hardware Validation** | *PCB Design, Power Electronics, Firmware Bring-up, Debugging*

- Took the board from requirements to an order-ready design in **two weeks and five revisions**, cutting BOM cost 20%
- Prepared it for **JLCPCB assembly** (DFM fixes, IPC Class 2); assembled boards in hand 2 weeks after ordering
- Wrote **ESP-IDF / FreeRTOS** bring-up firmware with host-side unit tests and a 1 kHz data logger, then ran a **1-hour full-system stress test** with zero errors
- Found **what Rev1 got wrong** from the logs: Pi under-voltage traced to 0.2 Ω of cable loss, and $I^2C$ failing at 400 kHz under load - now in a **Rev2 change list ready to build**

**Mars Rover Prototype** | **PCB Designs** | *Embedded Systems, Wireless, Power Electronics, PCB, System Integration*

- **IRC 2025 (electrical junior):** designed a custom **LoRa** RF link (SX1278) for 1 km bidirectional telemetry - chosen over off-the-shelf RC to carry mixed data (arm position, science feedback, encoder states) on a noise-resistant channel
- Wrote firmware across 4 **ESP32** nodes using a shared data stream on a single UART channel
- **IRC 2026 (Electrical Head):** Designed and iteratively built the rover PCB suite (images) including:
  - **200A / 24V / 4.8 kW, 6 rail** power distribution board with stackable motor driver slots and per-rail current sensing
  - motor drivers using half H-Bridge ICs, to be easily repairable
  - **STM32F103/STM32H523** development boards tailored per node (LoRa, drivetrain, robotic arm, sensors)
- Designed breakout boards adapting commercial devboards to rover pinouts, all with protection circuitry

## Experience

**Intuitive Evolution** | *PCB & Electronics Engineering Intern* *Sep 2026 – present*

- Responsible for the electrical design and firmware of the robot's controller board, from requirements and parts sourcing through fabrication, bring-up and testing

**Team Vyadh** | *Electrical Head* *July 2025 – July 2026*

- Led a 6-member electrical domain within a 33-member rover team - set delivery timelines, managed budget and inventory, and mentored 3 juniors to where they can independently handle most technical work
- Sourced components across 5+ vendors, balancing specs, availability, and cost rather than the cheapest
- Migrated all rover nodes to STM32 and implemented CAN across 6 nodes for noise immunity and hot-swap capability in a high-EMI drivetrain
- Defined domain-wide PCB standards - mandatory protection circuitry and standardized board dimensions for panel sharing to reduce fabrication cost

**Antropi Robotics, Bengaluru** | *Electronics & Firmware Intern* *May 2026 – July 2026*

- Designed and deployed **ESP32-S3** conveyor-control firmware using **ESP-IDF**, **FreeRTOS**, and **MQTT** - built a job-based state machine handling tray sensor logic, timeouts, and status feedback across 4 conveyor nodes
- Designed and fabricated 2 iterations of the **conveyor controller PCB**, revising the sensor interface and board-level fixes after a sensor change
- Built reusable **PID** motor-control firmware and a tuning GUI, commissioning position control for an autodoor

**Vanderlande India Pvt. Ltd., Noida** | *PLC Controls Intern* *May 2025 – June 2025*

- Programmed PLCs and built HMI interfaces for baggage handling systems using **Siemens S7-1500** and **TIA Portal**
- Designed a PLC-controlled conveyor system with modular Function Blocks and validated real-time control logic using FactoryIO and **S7-PLCSIM** (GitHub Repository)